

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

DLA

Control of Blood Pressure



St. George's University
Grenada, West Indies



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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0339	Discuss rapid control of blood pressure by baroreceptor reflex
SOM.MK.II.BPM1.3.CPR.2.PHYS.0340	Construct a step-by-step flow chart to indicate changes in the cardiovascular system that occur when the blood pressure changes in the body, taking examples of increase and a decrease in blood pressure.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0341	Discuss long-term regulation of blood pressure by Renin-angiotensin-aldosterone mechanism
SOM.MK.II.BPM1.3.CPR.2.PHYS.0342	Discuss the role of ADH (vasopressin) in control of blood pressure
SOM.MK.II.BPM1.3.CPR.2.PHYS.0343	Discuss other mechanisms of control of blood pressure such as low pressure baroreceptors and Atrial natriuretic peptide.



Recommended reading

1. Adapted from Geeky Medics, Regulation of Blood Pressure;
<https://geekymedics.com/regulation-of-blood-pressure/>
2. Costanzo LS. Physiology- 6 Edition. Published in 2018. Available from Elsevier.
3. Hall JE. Guyton and Hall Textbook of Medical Physiology – 13 Edition. Published in 2016. Available from Elsevier Inc

Key points

- **Blood pressure regulation** is a complex process, regulated by several mechanisms that work in unison to maintain **homeostasis**.
- **Rapid adjustments** in blood pressure are typically neurally mediated by the **baroreceptor reflex**.
- **Intermediate** and **long-term** regulation of blood pressure is predominantly mediated by **vasoactive compounds**.

Basics of blood pressure

- arteries, venules, capillaries, interstitium and lymphatics

Blood pressure (BP) readings give the clinician insight on myocardia function. BP is controlled by a variety of complex physiological mechanisms which allow both short term adaptation and longer-term maintenance of BP within a normal range. Blood pressure which is too high or too low can lead to a wide range of pathology (e.g. ruptured blood vessels, reduced perfusion to organs) and therefore the mechanisms which maintain BP homeostasis need to be robust.

Blood pressure can be measured in a number of different ways, with the most common being **systolic** and **diastolic** blood pressure:

- **Systolic blood pressure** (SBP) represents the pressure in the blood vessels when the heart contracts (systole).
- **Diastolic blood pressure** (DBP) represents the pressure in the blood vessels between heartbeats (diastole).
- **Mean arterial blood pressure** (MABP) is another method of assessing blood pressure.
- Mean arterial blood pressure = Cardiac output x Systemic vascular resistance

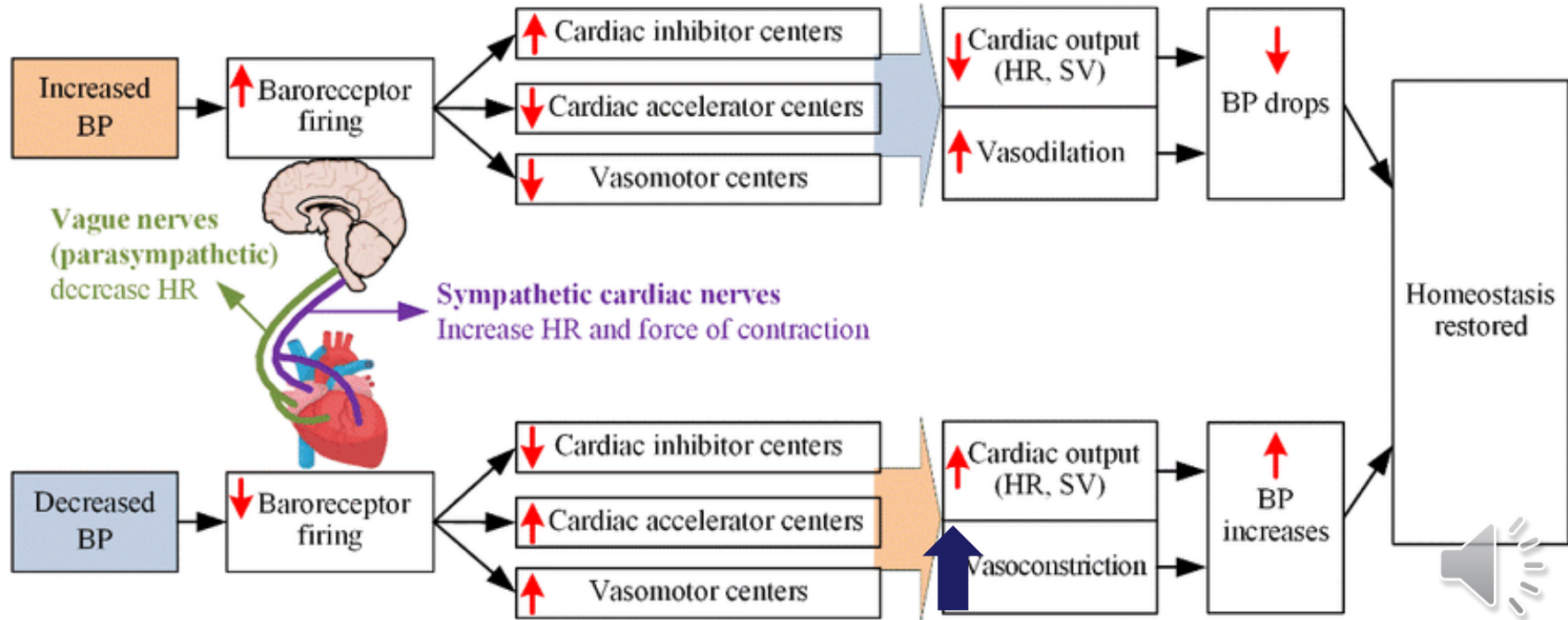
MAP can be calculated from the SBP and DBP using the following formula:

- $MABP = DBP + (Pulse\ pressure / 3)$
- Pulse pressure is calculated by subtracting DBP from SBP (i.e. $SBP - DBP$)



Rapid/ short-term control of blood pressure

The Baroreceptor reflex



The baroreceptor reflex

The **baroreceptor reflex** is a neurally-mediated reflex that regulates blood pressure in the short-term. This reflex is crucial for the maintenance of blood pressure throughout the day and in its absence, even a slight change in posture could lead to significant changes in blood pressure.

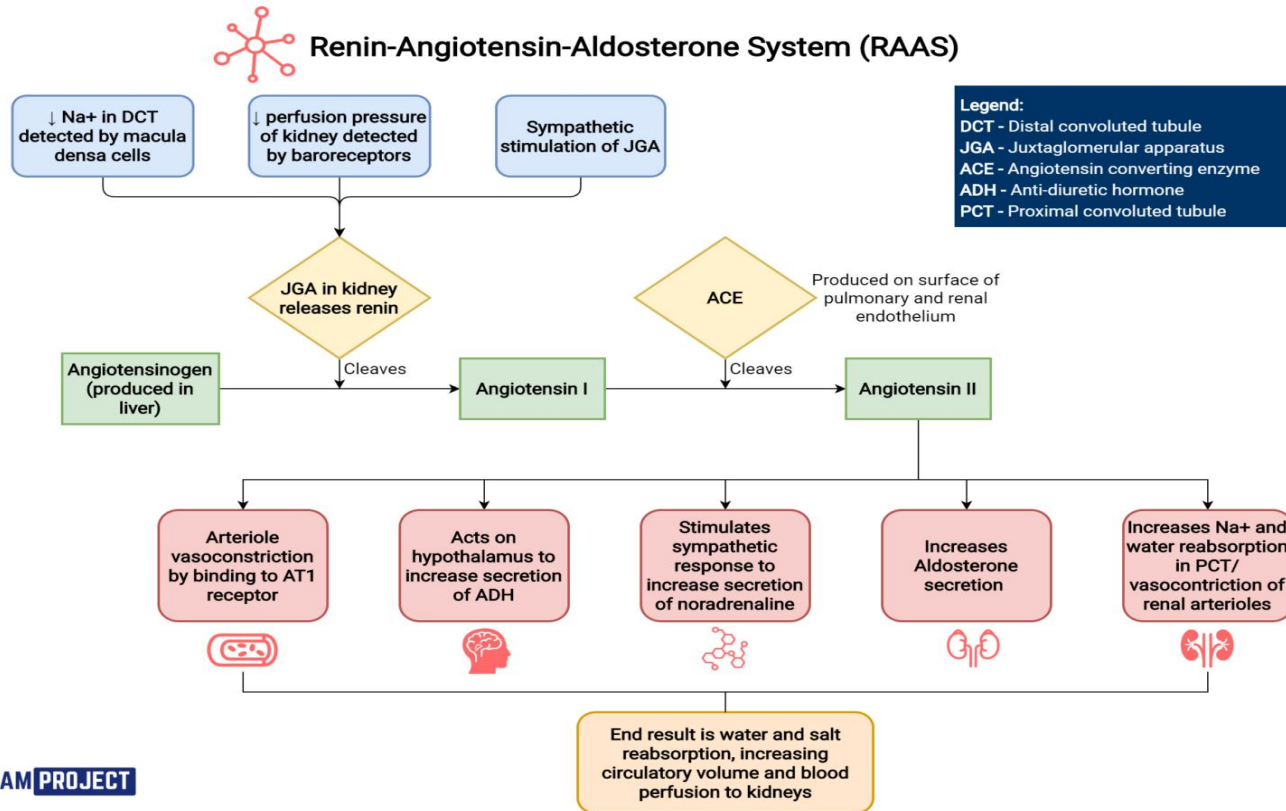
There are mechanoreceptors known as **baroreceptors** located in the **aortic arch** and **carotid sinus**, which constantly monitor the MABP and pulse pressure.¹ Increases in arterial pressure result in increased baroreceptor activity, which then increases the firing rate in the associated afferent neurons, carrying this information to the cardiovascular centre in the medulla.^{1,2}

In response to this, the parasympathetic nervous system (PSNS) activity increases and the sympathetic nervous system (SNS) activity decreases. The net effect of this is a reduction in heart rate and systemic vasodilatation collectively reducing MABP.

Conversely, when blood pressure decreases, there is less baroreceptor activation meaning the impulse rate of the afferent fibres decreases, causing the cardiovascular centre to increase SNS outflow and decrease PSNS outflow.² The net effect of this is increased cardiac contractility, increased heart rate and increased systemic vasoconstriction, causing an overall increase in blood pressure.^{1,2}

Intermediate and long-term regulation of blood pressure

Renin-angiotensin-aldosterone system (RAAS)



Renin-angiotensin-aldosterone system

The **renin-angiotensin-aldosterone system** (RAAS) is an essential component of blood pressure regulation that acts to increase blood volume and increase systemic vascular resistance. This system is dependent on hormonal changes which induce transcription of genes to produce vasoactive proteins, making it a slower means of controlling blood pressure compared to the baroreceptor reflex.

The RAAS system starts with **renin**, a hormone released from **granular cells** in the juxtaglomerular apparatus, a specialised structure involving parts of the distal collecting tubule (DCT) and the adjacent afferent arteriole of the glomerulus. Renin is released in response to increased concentration of salt in the blood, reduction in renal blood flow, or stimulation from the sympathetic nervous system acting on beta-1 receptors.

Renin converts **angiotensinogen**, a protein synthesised by the liver, into **angiotensinogen I**, which is subsequently converted by **angiotensinogen-converting enzyme** (ACE) into **angiotensin II**. Angiotensin II causes **vasoconstriction** in the systemic circulation and the renal microvasculature, preferentially constricting the efferent arteriole.

ACE, which is found primarily in the lungs, also rids the body of a vasodilator called bradykinin, causing further vasoconstriction. Importantly, angiotensin II functions to increase salt reabsorption at the level of the kidney and does so indirectly through the activation of **aldosterone** released from the zona glomerulosa of the adrenal cortex. Increased salt retention subsequently increases plasma volume and blood pressure.

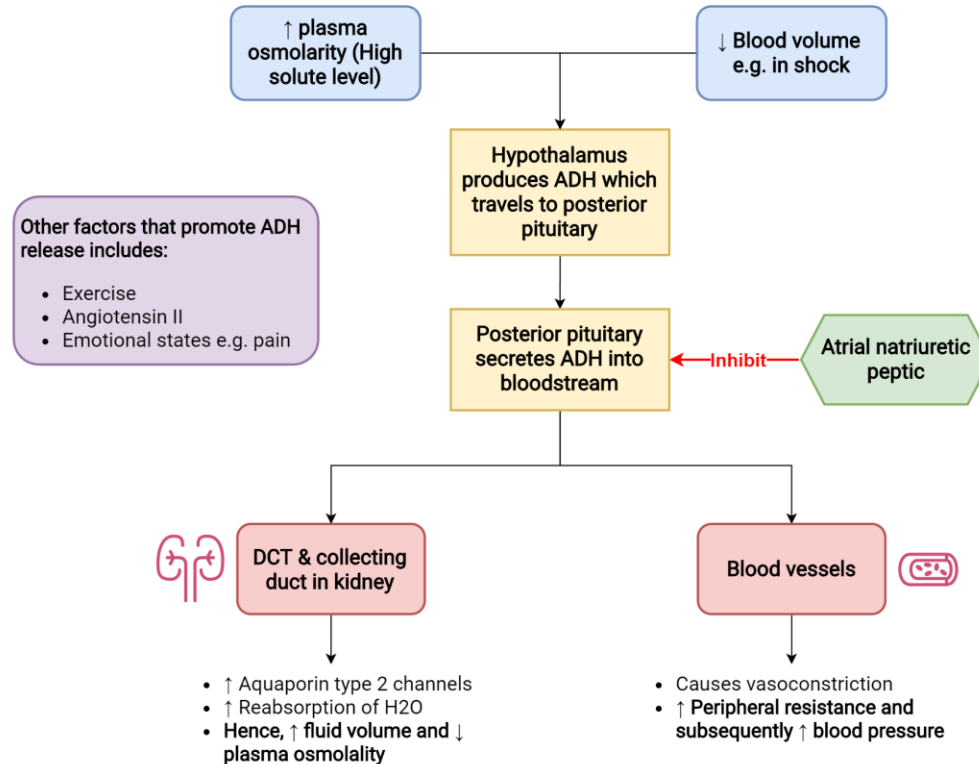
Angiotensin II is also capable of increasing plasma volume through the stimulation of thirst and antidiuretic hormone (ADH), another regulator of blood pressure which will be discussed shortly.

Aldosterone acts on the principal cells found in the DCT and collecting duct of the nephron, increasing Na⁺ reabsorption while simultaneously increasing K⁺ secretion into the tubules. Aldosterone-mediated salt resorption is also linked to H⁺ secretion. Given the ability of aldosterone to increase the volume of the extra-cellular fluid compartment and thus BP, several common anti-hypertensive medications aim to decrease blood pressure through inhibition of aldosterone formation.

Antidiuretic hormone (ADH)



Antidiuretic hormone (ADH)/ Arginine Vasopressin



<https://gramproject.com/diagram/antidiuretic-hormone-adh-vasopressin/>

Antidiuretic hormone (ADH)

Antidiuretic hormone, also known as vasopressin, is involved in the control of blood pressure. ADH is made by cell bodies located in the hypothalamus and released from the adjacent posterior pituitary. The following physiological changes trigger ADH release:

- an increase in plasma osmolarity (detected by osmoreceptors in the hypothalamus)
- a reduction in blood volume
- an increase in the levels of angiotensin II

ADH acts to increase water reabsorption by binding to V2 receptors, subsequently anchoring water channels known as **aquaporins** to the apical membrane of its target, principal cells in the collecting duct and DCT of the kidney. These aquaporins, named AQP-2 channels, are accountable for the variable H₂O permeability at the distal part of the nephron, seeing as water cannot pass through without them.

When someone becomes dehydrated, the osmolarity of the extracellular fluid increases, leading to ADH release from the posterior pituitary. Water is then reabsorbed at an increased rate at the level of the kidney, ultimately acting to increase the intravascular fluid volume. This increases blood pressure through an increase in venous pressure, thereby boosting venous return to the heart, increasing cardiac output.

ADH also acts as a vasoconstrictor targeting V1 receptors on vascular smooth muscle at high concentrations such as those seen in response to hemorrhagic shock.

Other regulators of blood pressure

Other regulators of blood pressure

Low-pressure baroreceptors

Low-pressure baroreceptors, in contrast to the high-pressure baroreceptors discussed previously, are found in the venous system, atria and pulmonary arteries. They respond to changes in plasma volume modulating blood pressure via a variety of mechanisms.

Atrial natriuretic peptide

Atrial natriuretic peptide (ANP) is a vasoactive peptide released from the atria in response to a rise in atrial pressures, which in turn are linked to venous pressure. ANP acts to lower blood pressure, primarily by vasodilation and the inhibition of sodium reabsorption by the kidney, the latter having a diuretic effect. This system increases sodium excretion in part through the opposition of the renin-angiotensin-aldosterone system, inhibiting renin and aldosterone release. ANP has also been shown to have inhibitory effects on vasopressin.



Summary of vasoactive compounds

Vasoactive compound	Site of production	Effects on vasculature	Effects on extracellular fluid (ECF) levels
Angiotensin II	Various sites: ACE gives rise to angiotensin II (mostly in the lungs)	Vasoconstriction	Increases ECF volume Increases sodium reabsorption independently, and stimulates aldosterone and ADH production
Aldosterone	Adrenal glands (cortex)	–	Increases ECF volume by increasing sodium reabsorption
Anti-diuretic hormone (ADH)	Hypothalamus (released from the posterior pituitary)	Vasoconstriction	Increases ECF volume by increasing H ₂ O reabsorption
Anti-natriuretic peptide (ANP)	Cardiomyocytes (responds to stretch of atria)	Vasodilatation	Decreases ECF volume by reducing sodium reabsorption

How do vasoactive compounds alter SVR/TPR and BP?

Vasoactive compounds often modify the amount of resistance in the systemic circulation (systemic vascular resistance – SVR) by targeting arterioles, the smallest arterial vessels.

The smooth muscle in these vessels contains several receptors, which when bound to, give rise to either of the following responses, depending on receptor type:

- **Stimulation of smooth muscle contraction:** decreasing vessel diameter and increasing systemic vascular resistance
- **Inhibition of smooth muscle:** increasing vessel diameter and reducing systemic vascular resistance

Changes to the diameter of these small vessels occur throughout the body, increasing the arteriolar tone. When the area through which blood passes decreases, blood pressure increases.



Practice MCQs

A 50-year-old man comes to the physician for a follow-up visit. His blood pressure was 146/92 mm Hg at his last office visit 4 weeks ago. He was diagnosed with essential hypertension 1 year ago. Laboratory evaluation at that time showed normal blood count, chemistry panel, and urinalysis. The patient claims that he exercises regularly and has been compliant with a low-salt diet. He has been taking a thiazide diuretic and angiotensin-converting enzyme inhibitor for hypertension. He quit smoking 2 years ago and has a 25-pack-year history. The patient drinks 3-4 glasses of wine every day and 6-8 beers on weekends with his friends. He does not use illicit drugs. His father died of a stroke at age 60. Today, the patient's blood pressure is 150/90 mm Hg and pulse is 76/min. The rest of his physical examination is normal. What is the most appropriate next step in management of this patient's hypertension?

- A. Admit to the hospital for blood pressure control
- B. Counsel for reduction in alcohol intake
- C. Administer antidiuretic hormone - vasopressin
- D. Obtain plasma aldosterone/renin ratio and urine metanephrines
- E. Start a beta blocker

A middle-aged man was recently diagnosed with hypertension. The prescribed antihypertensive medication enhances natriuresis, decreases serum angiotensin II concentration, and decreases aldosterone production. This medication most likely belongs to which of the following drug classes?

- A. Aldosterone receptor antagonist
- B. Alpha-adrenergic blocker
- C. Angiotensin II receptor blocker
- D. Direct renin inhibitor
- E. Loop diuretic

A 32-year-old man with diabetes presents to his physician with orthostatic hypotension. This suggests a deficiency in the normal physiologic response carried out by arterial baroreceptors located in the aortic arch and the carotid sinus. What is the normal physiologic response to hypotension?

- A. Decreased baroreceptor afferent firing in the aortic arch leads to increased sympathetic efferent firing
- B. Decreased baroreceptor afferent firing in the carotid sinus leads to increased parasympathetic efferent firing
- C. Decreased baroreceptor afferent firing in the carotid sinus leads to increased sympathetic efferent firing
- D. Increased baroreceptor afferent firing in the aortic arch leads to increased parasympathetic efferent firing
- E. Increased baroreceptor afferent firing in the carotid sinus leads to increased parasympathetic efferent firing



Practice MCQs – answers

A 50-year-old man comes to the physician for a follow-up visit. His blood pressure was 146/92 mm Hg at his last office visit 4 weeks ago. He was diagnosed with essential hypertension 1 year ago. Laboratory evaluation at that time showed normal blood count, chemistry panel, and urinalysis. The patient claims that he exercises regularly and has been compliant with a low-salt diet. He has been taking a thiazide diuretic and angiotensin-converting enzyme inhibitor for hypertension. He quit smoking 2 years ago and has a 25-pack-year history. The patient drinks 3-4 glasses of wine every day and 6-8 beers on weekends with his friends. He does not use illicit drugs. His father died of a stroke at age 60. Today, the patient's blood pressure is 150/90 mm Hg and pulse is 76/min. The rest of his physical examination is normal. What is the most appropriate next step in management of this patient's hypertension?

- A. Admit to the hospital for blood pressure control
- B. **Counsel for reduction in alcohol intake – lifestyle modification and reduction of risks**
- C. Administer antidiuretic hormone - vasopressin
- D. Obtain plasma aldosterone/renin ratio and urine metanephrines
- E. Start a beta blocker

A middle-aged man was recently diagnosed with hypertension. The prescribed antihypertensive medication enhances natriuresis, decreases serum angiotensin II concentration, and decreases aldosterone production. This medication most likely belongs to which of the following drug classes?

- A. Aldosterone receptor antagonist
- B. Alpha-adrenergic blocker
- C. Angiotensin II receptor blocker
- D. **Direct renin inhibitor**
- E. Loop diuretic

Disturbances in the renin-angiotensin-aldosterone system (RAAS) can play a significant role in development of hypertension. This patient's medication increases natriuresis, decreases serum angiotensin II concentration, and decreases aldosterone production. Of the listed medications, only direct renin inhibitors [D] (eg, aliskiren) cause these changes. (Choice A) Aldosterone receptor antagonists do not decrease angiotensin II concentration, but they do induce natriuresis and increase aldosterone levels by blocking the mineralocorticoid receptors. (Choice B) Alpha-adrenergic blockers are direct vasodilators used to treat hypertension. They decrease blood pressure but do not decrease angiotensin II or aldosterone concentrations or induce natriuresis. (Choice C) Angiotensin receptor blockers prevent angiotensin II from acting on angiotensin receptors. Unlike ACE inhibitors, angiotensin receptor blockers do not decrease angiotensin II levels but do cause natriuresis and decreased aldosterone production. (Choice E) Loop diuretics do induce natriuresis, but the decreased blood volume stimulates renin release that in turn increases angiotensin II and aldosterone concentrations.

A 32-year-old man with diabetes presents to his physician with orthostatic hypotension. This suggests a deficiency in the normal physiologic response carried out by arterial baroreceptors located in the aortic arch and the carotid sinus. What is the normal physiologic response to hypotension?

- A. Decreased baroreceptor afferent firing in the aortic arch leads to increased sympathetic efferent firing
- B. Decreased baroreceptor afferent firing in the carotid sinus leads to increased parasympathetic efferent firing
- C. **Decreased baroreceptor afferent firing in the carotid sinus leads to increased sympathetic efferent firing**
- D. Increased baroreceptor afferent firing in the aortic arch leads to increased parasympathetic efferent firing
- E. Increased baroreceptor afferent firing in the carotid sinus leads to increased parasympathetic efferent firing

The correct answer is C. The carotid sinus baroreceptor sends an afferent signal via the glossopharyngeal nerve to the medulla, which in turn responds with a sympathetic efferent signal that causes vasoconstriction, increased heart rate, increased contractility, and increased blood pressure. Answer B is incorrect. The baroreceptor located in the aortic arch responds only to an increase in blood pressure. Answer A is incorrect. **The correct efferent response to a decreased baroreceptor afferent firing rate would be increased sympathetic activity and decreased parasympathetic activity.** Answer D is incorrect. The baroreceptor located in the aortic arch responds only to an increase in blood pressure. Answer E is incorrect. The afferent firing rate would decrease, not increase, with hypotension.



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Any Questions? Come by my office, meet on zoom or email!

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